# Oluwajolasun Jaiyesimi

Calgary, Alberta. T2H 1P2 | (No work sponsorship required)
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## SKILLS

*Languages & Frameworks:* Python, JavaScript, TypeScript, React, FlaskAPI, Node.js, SQL (PostgreSQL), NoSQL (Elasticsearch).
*Tools & Platforms:* GCP, AWS, Pytest, Docker, Git, GitHub, GitHub Actions, Honeycomb, Pub/Sub, Postman, Shortcut, Jira, Trello.
*Technical & Soft Skills:* Full-Stack Development, REST APIs, Webhooks, Event-Driven Architectures, ETL Pipelines, Microservices, Machine Learning (TensorFlow, Keras, Hugging Face), Unit & Automated Testing, CI/CD, Agile/Scrum, Software Development Lifecycle, Design Patterns, Data Visualization, Spatial Data Analysis, Cross-functional Collaboration.

## WORK EXPERIENCE

**Areto Labs** **Remote, Canada.**
Software Engineer - Full Stack Web Developer July 2025 – Present

- Designed and deployed a microservice on GCP for detecting porn-bots on TikTok and Instagram, leveraging a third-party library for profile scraping and increasing detection catch rate by 60%.
- Trained and fine-tuned a Hugging Face model using internally collected datasets, including scraped profile photos, significantly boosting model accuracy and reducing false positives.
- Architected event-driven data ingestion systems using Google Pub/Sub to stream topic and subscription messages between microservices, leveraging Honeycomb traces to monitor, debug, and optimize event flows.
- Implemented and maintained DevOps pipelines using GCP, GitHub, and GitHub Actions, improving deployment speed and reliability.
- Collaborated in code reviews, architectural discussions, and team strategy, while ensuring software quality with unit, integration, and end-to-end testing.
- Optimized React and TypeScript front-end applications to deliver seamless, user-focused experiences.

**CANN Forecast** **Montreal, Quebec.**
Software Engineer - Full Stack Web Developer September 2023 – September 2024

- Managed PostgreSQL databases, built ETL pipelines with Pandas, and performed spatial analysis using QGIS and GeoPandas for over 10 water infrastructure projects.
- Collaborated with an agile team of domain experts to develop React based interfaces, interactive maps using Leaflet.js and Mapbox, and visualizations with D3.js, enhancing accessibility and enabling data-driven decisions for 100+ users.
- Automated email attachment processing via Python (IMAP/SMTP), cutting manual work by 90%, and integrated the solution with Jenkins, resulting in faster deployment cycles and reduced processing time by 4 hours per day.
- Engineered scalable backend systems in Django, exposing RESTful APIs that supported dynamic front-end functionality, serving over 500 concurrent users.

**Hifi Engineering** **Calgary, Alberta.**
Software Engineer - Machine Learning Engineer September 2022 – July 2023

- Designed and implemented ML models for sensor data analysis, achieving 95% accuracy in simulated pipeline leak detection using Keras and TensorFlow.
- Reduced false positives by 15% through close collaboration with domain experts during model tuning and optimization.
- Improved system sensitivity by 20% via rigorous validation, significantly enhancing real-world reliability.
- Communicated technical results to both technical and non-technical stakeholders, contributing to a projected 30% reduction in operational risk.

## EDUCATION

**Master's Degree, Software Engineering (MEng)** **May 2021 – June 2023**
University of Calgary Calgary, Alberta.

- GPA: 3.85 / 4.00

**Bachelor's Degree, Electrical and Electronics Engineering (BEng)** **October 2015 – July 2018**
Loughborough University Loughborough, United Kingdom.

- 2nd Class Upper (GPA Equivalent: 3.72 / 4.00)

## TECHNICAL PROJECTS

Email Attachment Uploader/Downloader: https://github.com/oluwajolasun/Email-Automation-for-Client-Data

- Automated attachment handling using Python's IMAP/SMTP libraries, cutting 90% of manual effort.
- Integrated with Jenkins for seamless deployment and CI/CD workflows.

Novel CNN for Fish Classification (FishNet): https://github.com/oluwajolasun/ENEL645_FinalProject_FishClassification

- Designed a custom CNN ("Fishnet") achieving 99.3% test accuracy with regularization techniques.
- Benchmarked against VGG16 and EfficientNetB1 using transfer learning on a 9,430-image dataset.